

Computer Networks



Overview of the Internet Protocol layering (Book 1 Ch 1)

Application Layer

- Application-Layer Paradigms
- Client-server applications
 - World Wide Web and HTTP
 - FTP
 - Electronic Mail
 - DNS
- Peer-to-peer paradigm
 - P2P Networks
- Case study: BitTorrent (Book 1 Ch 2)

① Introduction

- Providing Services
- Application-Layer Paradigms

② Standard Client-Server Applications

- World Wide Web and HTTP
- FTP
- Electronic Mail
- Domain Name System (DNS)

③ Peer-to-Peer Paradigm

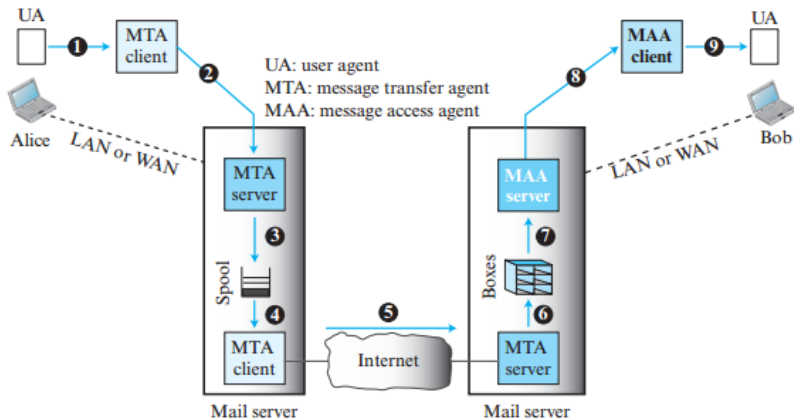
- P2P Networks

Electronic Mail (E-Mail)

- Electronic mail allows:
 - Exchange of messages between users
- E-mail differs from HTTP and FTP because:
 - It is mainly a one-way transaction
- Example:
 - Alice sends mail to Bob
 - Bob may or may not reply
- Each reply forms:
 - Another independent transaction
- Users do not run mail servers continuously:
 - Computers may be turned off

Common scenario

Figure 2.19 Common scenario



Architecture of E-Mail System

- E-mail system uses:
 - Intermediate mail servers
- Sender and receiver connected through:
 - LAN or WAN
- Mail server provides:
 - Mailboxes
 - Queue (spool)
- Mailbox:
 - Stores received messages
- Queue (spool):
 - Stores outgoing messages waiting to be sent



Components of E-Mail System

- E-mail system uses:
 - User Agent (UA)
 - Message Transfer Agent (MTA)
 - Message Access Agent (MAA)
- User Agent (UA):
 - Prepares and reads messages
- Message Transfer Agent (MTA):
 - Transfers messages between mail servers
- Message Access Agent (MAA):
 - Retrieves messages from mailbox

Steps in Sending and Receiving E-Mail

- Alice uses:
 - User Agent to prepare message
- Message stored in:
 - Queue at sender mail server
- MTA transfers message:
 - Across Internet
- Bob retrieves mail using:
 - Message Access Agent (MAA)
- MAA acts as:
 - Pull mechanism

Push and Pull Operations in E-Mail

- MTA operation:
 - Push mechanism
- Sender pushes message to:
 - Mail server
- Bob cannot directly use MTA server because:
 - System would need to stay online continuously
- MAA operation:
 - Pull mechanism
- Receiver pulls message from:
 - Mail server mailbox

User Agent (UA)

- User Agent provides:
 - User interface for e-mail
- Functions of UA:
 - Compose messages
 - Read messages
 - Reply to messages
 - Forward messages
 - Manage local mailbox
- Two types of UAs:
 - Command-driven
 - GUI-based

Types of User Agents

- Command-driven UAs:
 - Early e-mail programs
 - Use keyboard commands
- Examples:
 - mail
 - pine
 - elm
- GUI-based UAs:
 - Use graphical interface
 - Easier interaction using mouse and keyboard
- Examples:
 - Eudora
 - Outlook



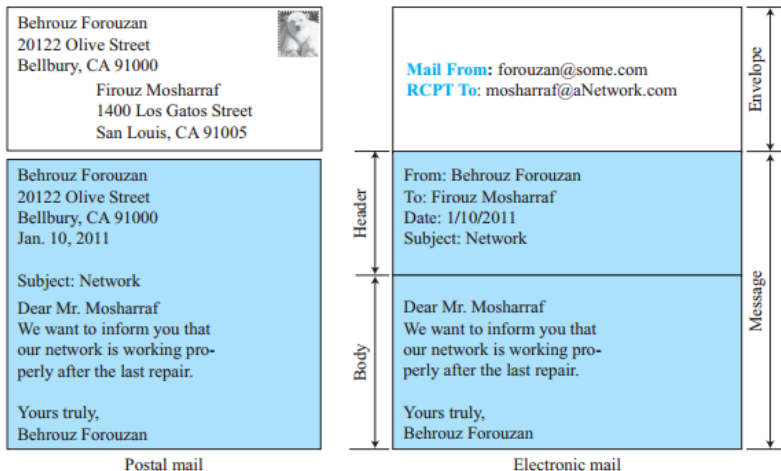
Sending and Receiving Mail

- E-mail contains:
 - Envelope
 - Message
- Envelope includes:
 - Sender address
 - Receiver address
- Message contains:
 - Header
 - Body
- Header defines:
 - Sender
 - Receiver
 - Subject
- Body contains:
 - Actual message content



Format of an e-mail

Figure 2.20 *Format of an e-mail*



Receiving Mail and Mail Addresses

- User Agent informs user:
 - When new mail arrives
- Message summary includes:
 - Sender address
 - Subject
 - Time of message
- E-mail address format:
 - local-part@domain-name
- Local part:
 - User mailbox name
- Domain name:
 - Mail server or organization name

Mailing List and SMTP

- Mailing list:
 - One alias represents multiple e-mail addresses
- System creates:
 - Separate message for each recipient
- SMTP stands for:
 - Simple Mail Transfer Protocol
- SMTP used between:
 - Sender and sender mail server
 - Two mail servers
- SMTP defines:
 - Exchange of commands and responses

E-Mail Addresses

- Mail system requires:
 - Unique addressing system
- Internet e-mail address contains:
 - Local part
 - Domain name
- Two parts separated by:
 - @ symbol
- General format:

`local-part@domain-name`

- Example:

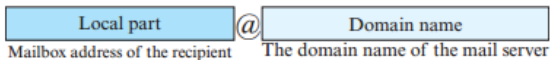
`alice@company.com`

Local Part and Domain Name

- Local part defines:
 - User mailbox name
- User mailbox:
 - Special file storing received e-mails
- Messages retrieved using:
 - Message Access Agent (MAA)
- Domain name identifies:
 - Mail server or organization
- Mail servers also called:
 - Mail exchangers
- Domain names obtained from:
 - DNS database
 - Logical organization names

E-mail address

Figure 2.21 *E-mail address*

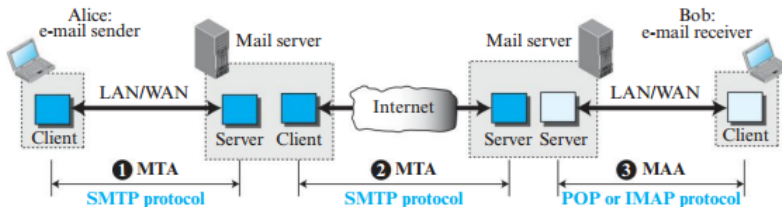


Mailing List or Group List

- Electronic mail supports:
 - Mailing lists (group lists)
- Mailing list:
 - One alias represents multiple e-mail addresses
- Before sending mail:
 - System checks alias database
- If alias corresponds to mailing list:
 - Separate message created for each member
- Messages then handed to:
 - Message Transfer Agent (MTA)

Protocols used in electronic mail

Figure 2.22 *Protocols used in electronic mail*



Message Transfer Agent (MTA) and SMTP

- E-mail system uses:
 - Three client-server applications
- First two applications:
 - Message Transfer Agents (MTAs)
- Third application:
 - Message Access Agent (MAA)
- SMTP stands for:
 - Simple Mail Transfer Protocol
- SMTP defines:
 - Communication between MTA client and MTA server

Use of SMTP

- SMTP used in two places:
 - Between sender and sender's mail server
 - Between two mail servers
- Another protocol required:
 - Between receiver and mail server
- SMTP operation based on:
 - Commands
 - Responses
- SMTP specifies:
 - Format of command and response exchange

SMTP Commands and Responses

- SMTP uses:
 - Commands from client to server
 - Responses from server to client
- Each command or response terminated by:
 - Carriage Return (CR)
 - Line Feed (LF)
- Command format:
 - Keyword
 - Zero or more arguments
- SMTP defines:
 - 14 commands
- Response format:
 - Three-digit code
 - Optional textual information

SMTP Commands

Table 2.6 SMTP Commands

Keyword	Argument(s)	Description
HELO	Sender's host name	Identifies itself
MAIL FROM	Sender of the message	Identifies the sender of the message
RCPT TO	Intended recipient	Identifies the recipient of the message
DATA	Body of the mail	Sends the actual message
QUIT		Terminates the message

Table 2.6 SMTP Commands (continued)

Keyword	Argument(s)	Description
RSET		Aborts the current mail transaction
VERFY	Name of recipient	Verifies the address of the recipient
NOOP		Checks the status of the recipient
TURN		Switches the sender and the recipient
EXPN	Mailing list	Asks the recipient to expand the mailing list.
HELP	Command name	Asks the recipient to send information about the command sent as the argument
SEND FROM	Intended recipient	Specifies that the mail be delivered only to the terminal of the recipient, and not to the mailbox
SMOL FROM	Intended recipient	Specifies that the mail be delivered to the terminal <i>or</i> the mailbox of the recipient
SMAL FROM	Intended recipient	Specifies that the mail be delivered to the terminal <i>and</i> the mailbox of the recipient

SMTP Responses

Table 2.7 Responses

Code	Description
Positive Completion Reply	
211	System status or help reply
214	Help message
220	Service ready
221	Service closing transmission channel
250	Request command completed
251	User not local; the message will be forwarded
Positive Intermediate Reply	
354	Start mail input
Transient Negative Completion Reply	
421	Service not available
450	Mailbox not available
451	Command aborted: local error
452	Command aborted; insufficient storage
Permanent Negative Completion Reply	
500	Syntax error; unrecognized command
501	Syntax error in parameters or arguments
502	Command not implemented
503	Bad sequence of commands
504	Command temporarily not implemented
550	Command is not executed; mailbox unavailable
551	User not local
552	Requested action aborted; exceeded storage location
553	Requested action not taken; mailbox name not allowed
554	Transaction failed

Mail Transfer Phases in SMTP

- SMTP mail transfer occurs in:
 - Connection establishment
 - Message transfer
 - Connection termination
- SMTP uses:
 - TCP connection on port 25
- Commands and responses exchanged between:
 - SMTP client
 - SMTP server

Connection Establishment Phase

- After TCP connection established:
 - SMTP server starts communication
- Step 1:
 - Server sends code 220
 - Means "Service Ready"
- If service unavailable:
 - Server sends code 421
- Step 2:
 - Client sends HELO command
 - Includes client domain name
- Step 3:
 - Server responds with code 250
 - Means request completed

Message Transfer Phase

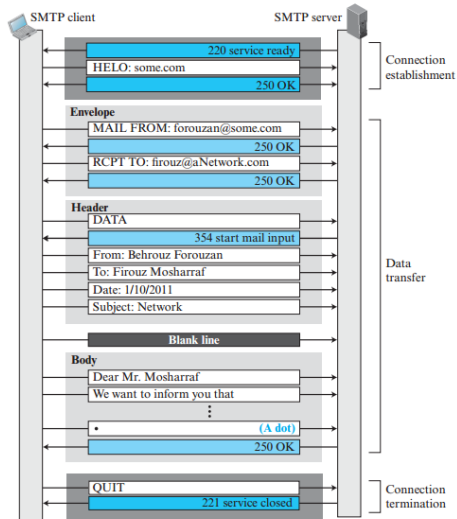
- Step 1:
 - Client sends MAIL FROM command
 - Contains sender address
- Step 2:
 - Server responds with code 250
- Step 3:
 - Client sends RCPT TO command
 - Contains recipient address
- Step 4:
 - Server responds with code 250
- RCPT TO steps repeated:
 - For multiple recipients

Message Transfer and Termination

- Step 5:
 - Client sends DATA command
- Step 6:
 - Server responds with code 354
 - Means “Start Mail Input”
- Step 7:
 - Client sends message contents
 - Each line ends with CR and LF
 - Message ends with single period
- Step 8:
 - Server responds with code 250
- Connection termination:
 - Client sends QUIT command
 - Server responds with code 221

Example

Figure 2.23 Example 2.13



Example 2.13: SMTP Mail Transfer

- Example demonstrates:
 - Three SMTP mail transfer phases
- Phases include:
 - Connection establishment
 - Message transfer
 - Connection termination
- Figure separates:
 - Envelope information
 - Header information
 - Message body

SMTP Message Transfer Process

- SMTP transfer steps repeated:
 - Two times for each e-mail
- First transfer:
 - Sender → Local mail server
- Second transfer:
 - Local mail server → Remote mail server
- SMTP used between:
 - Mail transfer agents (MTAs)

Role of the Local Mail Server

- Local mail server receives:
 - Complete e-mail message
- Message may be:
 - Stored temporarily in spool (queue)
- Local server can send message:
 - Immediately
 - At a later time
- Spooling helps:
 - Manage delayed transmission
 - Improve reliability

Message Access Agent: POP3 and IMAP4

- First two stages of e-mail delivery use:
 - SMTP
- SMTP is:
 - Push protocol
- Third stage requires:
 - Pull protocol
- Client must:
 - Retrieve messages from mail server
- Third stage uses:
 - Message Access Agent (MAA)
- Two common message access protocols:
 - POP3
 - IMAP4

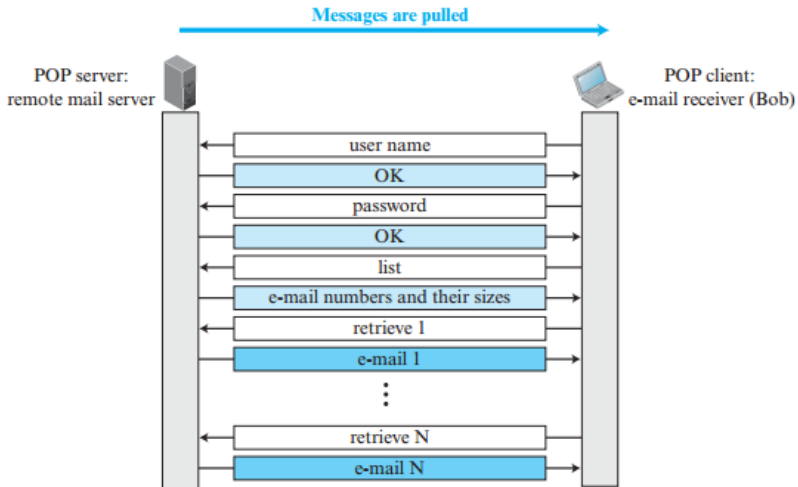


POP3 (Post Office Protocol Version 3)

- POP3:
 - Simple mail access protocol
- Client POP3 software:
 - Installed on recipient computer
- Server POP3 software:
 - Installed on mail server
- Mail access process:
 - Client opens TCP connection on port 110
 - Sends username and password
 - Retrieves messages one by one

POP3

Figure 2.24 POP3



POP3 Modes

- POP3 supports two modes:
 - Delete mode
 - Keep mode
- Delete mode:
 - Mail deleted after retrieval
 - Used on permanent computer
- Keep mode:
 - Mail remains on server after retrieval
 - Used for remote access or laptops
- Keep mode allows:
 - Later retrieval and organization

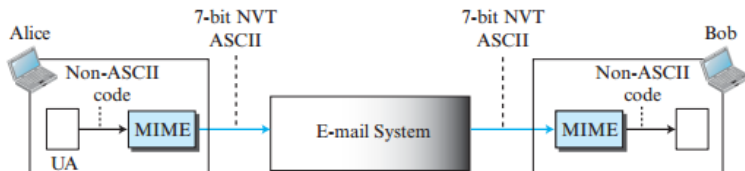
IMAP4 (Internet Mail Access Protocol Version 4)

- IMAP4:
 - More powerful than POP3
 - More complex than POP3
- POP3 limitations:
 - Cannot organize mail on server
 - Cannot partially check contents before download
- IMAP4 provides:
 - Header checking before download
 - Content searching before download
 - Partial download of e-mail

Additional Features of IMAP4

- IMAP4 supports:
 - Partial multimedia download
- Useful when:
 - Bandwidth is limited
- User can:
 - Create mailboxes
 - Delete mailboxes
 - Rename mailboxes
- IMAP4 also allows:
 - Hierarchy of folders/mailboxes
- Improves:
 - E-mail organization on server

Figure 2.25 *MIME*



MIME (Multipurpose Internet Mail Extensions)

- Electronic mail has:
 - Simple message structure
- Traditional e-mail supports only:
 - NVT 7-bit ASCII format
- Limitation:
 - Cannot support many non-English languages
- Examples of unsupported languages:
 - French
 - German
 - Russian
 - Chinese
 - Japanese
- Traditional e-mail also cannot send:
 - Binary files
 - Audio files
 - Video files

Purpose and Function of MIME

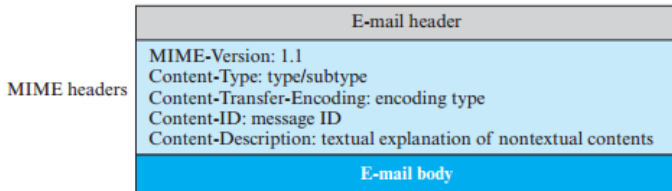
- MIME stands for:
 - Multipurpose Internet Mail Extensions
- MIME is:
 - Supplementary e-mail protocol
- MIME allows:
 - Transmission of non-ASCII data through e-mail
- At sender side:
 - Non-ASCII data converted to ASCII format
- Data sent through:
 - Standard e-mail system
- At receiver side:
 - ASCII data converted back to original form

Working of MIME

- MIME behaves like:
 - Set of software functions
- MIME performs:
 - ASCII to non-ASCII conversion
 - Non-ASCII to ASCII conversion
- MIME enables e-mail to support:
 - Multimedia data
 - International languages
 - Binary attachments
- MIME works together with:
 - Existing e-mail protocols

MIME Header

Figure 2.26 *MIME header*



MIME Headers

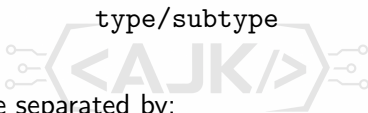
- MIME adds:
 - Extra headers to original e-mail header
- Purpose of MIME headers:
 - Define transformation parameters
- MIME defines:
 - Five additional headers
- Important MIME headers:
 - MIME-Version
 - Content-Type

MIME-Version Header

- MIME-Version header:
 - Specifies MIME version used
- Current MIME version:
 - Version 1.1
- Helps receiver:
 - Interpret MIME message correctly

Content-Type Header

- Content-Type header defines:
 - Type of data in message body
- Content-Type format:



- Type and subtype separated by:
 - Slash (/)
- Header may include:
 - Additional parameters
- MIME supports:
 - Seven different data types

Data Types and Subtypes in MIME

Table 2.8 *Data Types and Subtypes in MIME*

Type	Subtype	Description
Text	Plain	Unformatted
	HTML	HTML format (see Appendix C)
Multipart	Mixed	Body contains ordered parts of different data types
	Parallel	Same as above, but no order
	Digest	Similar to Mixed, but the default is message/RFC822
	Alternative	Parts are different versions of the same message
Message	RFC822	Body is an encapsulated message
	Partial	Body is a fragment of a bigger message
	External-Body	Body is a reference to another message
Image	JPEG	Image is in JPEG format
	GIF	Image is in GIF format
Video	MPEG	Video is in MPEG format
Audio	Basic	Single channel encoding of voice at 8 KHz
Application	PostScript	Adobe PostScript
	Octet-stream	General binary data (eight-bit bytes)

Content-Transfer-Encoding Header

- Content-Transfer-Encoding header:
 - Defines encoding method used in MIME
- Purpose:
 - Convert message into 0s and 1s for transport
- Encoding ensures:
 - Non-ASCII data can travel through e-mail systems
- MIME defines:
 - Five encoding methods
- Common encoding methods include:
 - 7bit
 - 8bit
 - Binary
 - Base64
 - Quoted-printable

Methods for Content-Transfer-Encoding

Table 2.9 *Methods for Content-Transfer-Encoding*

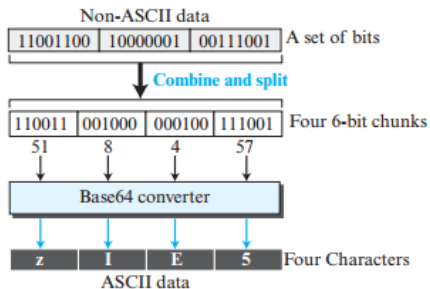
<i>Type</i>	<i>Description</i>
7-bit	NVT ASCII characters with each line less than 1000 characters
8-bit	Non-ASCII characters with each line less than 1000 characters
Binary	Non-ASCII characters with unlimited-length lines
Base64	6-bit blocks of data encoded into 8-bit ASCII characters
Quoted-printable	Non-ASCII characters encoded as an equal sign plus an ASCII code

Base64 Encoding

- Base64:
 - One of the MIME encoding methods
- Used for:
 - Encoding non-ASCII and binary data
- In Base64 encoding:
 - Data treated as stream of bits
- Bit stream divided into:
 - 6-bit chunks
- Each 6-bit group mapped to:
 - Printable ASCII character
- Base64 helps:
 - Safely transfer binary data through e-mail

Base64 conversion

Figure 2.27 Base64 conversion



Base64 Converting Table

Table 2.10 *Base64 Converting Table*

<i>Value</i>	<i>Code</i>	<i>Value</i>	<i>Code</i>	<i>Value</i>	<i>Code</i>	<i>Value</i>	<i>Code</i>	<i>Value</i>	<i>Code</i>	<i>Value</i>	<i>Code</i>
0	A	11	L	22	W	33	h	44	s	55	3
1	B	12	M	23	X	34	i	45	t	56	4
2	C	13	N	24	Y	35	j	46	u	57	5
3	D	14	O	25	Z	36	k	47	v	58	6
4	E	15	P	26	a	37	l	48	w	59	7
5	F	16	Q	27	b	38	m	49	x	60	8
6	G	17	R	28	c	39	n	50	y	61	9
7	H	18	S	29	d	40	o	51	z	62	+
8	I	19	T	30	e	41	p	52	0	63	/
9	J	20	U	31	f	42	q	53	1		
10	K	21	V	32	g	43	r	54	2		

Base64 Encoding Process

- In Base64 encoding:
 - Each 6-bit section converted to ASCII character
- Conversion performed using:
 - Base64 character table
- Base64 is:
 - Redundant encoding scheme
- Reason:
 - 6 bits converted into 8-bit ASCII character
- Encoding overhead:
 - 25%

Quoted-Printable Encoding

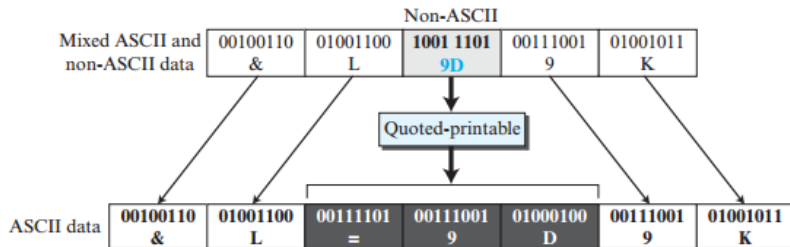
- Used when:
 - Data mostly contains ASCII characters
- ASCII characters:
 - Sent unchanged
- Non-ASCII characters:
 - Replaced using three ASCII characters
- Format:
 - Equal sign (=)
 - Two hexadecimal digits
- Example:
 - Non-ASCII byte $9D_{16}$ becomes =9D

Additional MIME Headers

- Content-ID header:
 - Uniquely identifies message
 - Used in multiple message environments
- Content-Description header:
 - Describes body content type
- Content-Description may indicate:
 - Image
 - Audio
 - Video

Quoted-printable

Figure 2.28 Quoted-printable



Web-Based Mail

- E-mail service provided through:
 - Websites on the Internet
- Popular web-based mail services:
 - Hotmail
 - Yahoo Mail
 - Google Mail (Gmail)
- Users can:
 - Access e-mail using web browser
- No separate mail software required:
 - Browser acts as user interface
- Web-based mail simplifies:
 - Sending and receiving e-mails

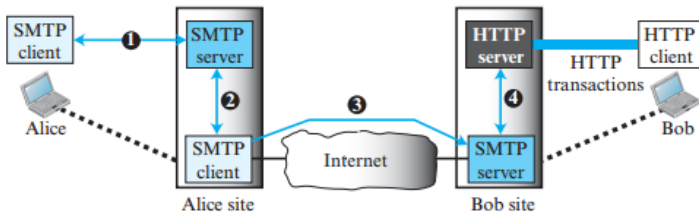


Working of Web-Based Mail

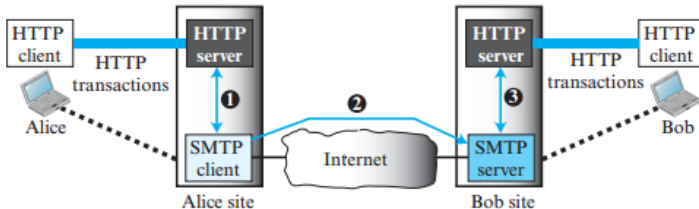
- User accesses:
 - Mail website through browser
- Website provides:
 - E-mail interface
 - Mail storage
 - Message transfer services
- Browser communicates with:
 - Web mail server
- User can:
 - Compose mail
 - Send mail
 - Read mail
 - Organize mail
- Entire service handled through:
 - Web applications

Web-based e-mail, cases I and II

Figure 2.29 Web-based e-mail, cases I and II



Case 1: Only receiver uses HTTP



Case 2: Both sender and receiver use HTTP

Web-Based Mail: Case I

- Alice uses:
 - Traditional mail server
- Bob uses:
 - Web-based mail server
- Mail transfer from Alice to her mail server:
 - Uses SMTP
- Transfer between mail servers:
 - Also uses SMTP
- Transfer from Bob's web server to browser:
 - Uses HTTP instead of POP3/IMAP4

Working of Case I

- Bob retrieves mail using:
 - HTTP requests
- Web server sends:
 - Login form
- Bob provides:
 - Username
 - Password
- After authentication:
 - List of e-mails sent in HTML format
- Bob can:
 - Browse and retrieve e-mails one by one



Web-Based Mail: Case II

- Both Alice and Bob use:
 - Web-based mail servers
- Alice sends mail using:
 - HTTP transactions
- Alice's browser sends:
 - HTTP request to web server
- Bob's mailbox address used as:
 - URL
- Web server passes message to:
 - SMTP client
- Transfer between mail servers:
 - Still uses SMTP
- Bob retrieves mail using:
 - HTTP transactions

- Basic e-mail protocols:
 - Do not provide built-in security
- Security mechanisms required for:
 - Confidentiality
 - Authentication
 - Data integrity
- E-mail security can be achieved using:
 - Application-layer security protocols
- Two important e-mail security protocols:
 - Pretty Good Privacy (PGP)
 - Secure/Multipurpose Internet Mail Extensions (S/MIME)
- These protocols provide:
 - Secure e-mail communication